

1.8 Quiz Corrections Solutions: Measurement

Notation: $=$ joins values that are exactly equal; $\approx$ introduces a rounded value. Each conversion runs as one chain of unit factors and is evaluated once.

1. Significant digits.

(a) **3**

(c) **2**

(e) **1**

(b) **3**

(d) **4**

(f) **3**

(a) Leading zeros do not count; the trailing zero after the decimal point does. (b) A zero between nonzero digits always counts. (d) Both zeros count: one sits between nonzero digits, the other trails after the decimal point. (e) 0.000 03 has a single measured digit, the 3. (f) In scientific notation, count the digits of the 1.67; the power of ten only places the decimal point.

2. Round to three significant figures.

(a) $\sqrt{7} = 2.6457513 \dots \approx$ **2.65**

(b) 4805.59 m $\approx$ **4810 m**, better written **4.81×10^3 m**

The third figure is the 0 in the tens place; the 5 after it rounds it up to 1. Written as 4810 m the final zero is only a placeholder, which is why scientific notation is the clearer answer.

3. Scientific notation.

(a) 149,597,871 km $= 1.49597871 \times 10^8$ km $\approx$ **1.50×10^8 km**

The third figure, 9, rounds up and carries: 1.495... becomes 1.50. Write 1.50, not 1.5 — the trailing zero is the third significant figure.

(b) 0.000 000 000 052 917 7 m $= 5.29177 \times 10^{-11}$ m $\approx$ **5.29×10^{-11} m**

Three groups of three zeros, then one more zero before the 5: ten zeros, so the 5 is in the 11th decimal place and the exponent is -11 .

4. Ordinary decimal numbers.

(a) 1.0×10^{-4} m = **0.000 10 m**

Keep the trailing zero: 1.0 has two significant figures and so must the decimal form. 0.0001 m loses one.

(b) 3.84×10^5 km = **384,000 km**

5. Reading the ruler. The bar ends between the 6.7 and 6.8 cm marks, a little short of halfway.

(a) **6.74 cm** — the 6.7 is certain and the final 4 is the estimated digit. *Accept 6.73 to 6.75.*

(b) **Three significant figures.**

(c) $6.74 \text{ cm} \times \frac{10 \text{ mm}}{1 \text{ cm}} = \mathbf{67.4 \text{ mm}}$

6. Convert.

(a) $828 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 2715.84 \text{ ft} \approx \mathbf{2720 \text{ ft}}$, or $\mathbf{2.72 \times 10^3 \text{ ft}}$

The published height is 2717 ft, so this checks.

(b) $300 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 83.333 \dots \approx \mathbf{83.3 \text{ m/s}}$

7. Skydiver.

$$55 \frac{\text{m}}{\text{s}} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 123 \text{ mi/h} \approx \mathbf{1.2 \times 10^2 \text{ mi/h}}$$

Two significant figures, set by “about 55 m/s”. Written as 120 mi/h the zero is only a placeholder, so scientific notation states the precision honestly. Accept 123 mi/h for full credit on the conversion; the sig-fig point is the one the quiz corrections are meant to practise.